

Experiment - 8.1.2. Prime Numbers in a Given Range

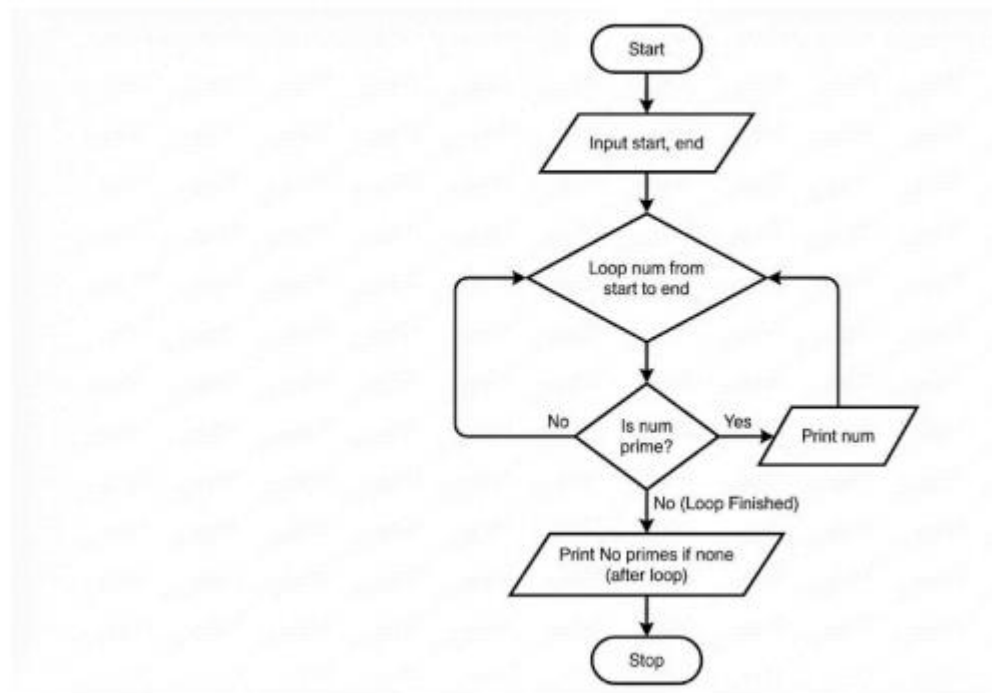
1. Aim

To design and implement a Python program that finds and prints all prime numbers within a specified range [start, end] (inclusive). The program determines the primality of each number and prints it. If no prime numbers exist within the given range, the program prints "No primes".

2. Pseudocode

1. **START**
2. **READ** an integer representing the start of the range and **STORE** it in start.
3. **READ** an integer representing the end of the range and **STORE** it in end.
4. **INITIALIZE** a boolean variable found to False to keep track of whether any primes exist.
5. **FOR** each number num in the range from start to end (inclusive):
 - **IF** num > 1 (since 1 and numbers below it are not prime):
 - **INITIALIZE** is_prime to True.
 - **FOR** each number i from 2 up to the integer square root of num ($\text{int}(\text{num}^{0.5}) + 1$):
 - **IF** num is divisible by i ($\text{num} \% i == 0$):
 - **SET** is_prime to False.
 - **BREAK** the inner loop.
 - **IF** is_prime is True:
 - **PRINT** the prime number num.
 - **SET** found to True.
6. **IF** found is False (meaning no primes were printed):
 - **PRINT** "No primes".
7. **END**

3. Flowchart



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8.1.2. Prime Numbers in a Given Range

Write a Python program to print all prime numbers within a given range (inclusive of both start and end values).

Input Format:

- First line contains an integer representing the start of the range.
- Second line contains an integer representing the end of the range.

Output Format:

- Print all prime numbers in the given range (including start and end if they are prime). Print each prime number on a separate line.
- If no prime numbers exist in the range, print: "No primes".

Sample Test Cases

primeNu...

```
1 st = int(input())
2 stop = int(input())
3 f = False
4 for n in range(st, stop + 1):
5     if n > 1:
6         flag = True
7         for i in range(2, n):
8             if n % i == 0:
9                 flag = False
10                break
11        if flag:
12            print(n)
13            f = True
14    if not f:
15        print("No primes")
```

TerminalTest cases

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